Madhav Trivedi

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LinkedIn | GitHub | Portfolio

SUMMARY

Software Engineer with **3+ years of experience** designing and building backend services, REST APIs and full-stack applications using C#, .NET, TypeScript, Angular, AWS and Azure. Delivered event-driven services handling 1,000 daily exports and 3,000 daily imports, shipped AI-powered features across the stack, and consolidated fragmented SDKs into a shared library used by 4 teams.

EXPERIENCE

Zeus Learning Jan 2024 – Present
Software Engineer *Mumbai, India*

- Engineered a fault-tolerant IMS QTI 2.2 export pipeline handling **1,000 packages/day** using .NET and RabbitMQ, cutting failures from **8% to 0.4%** and saving **~100 hours/month** through database checkpoints, Redis retries, and dead-letter queues that automatically recovered interrupted jobs without duplicate processing.
- Owned the QTI import pipeline end-to-end for **3,000 daily uploads**, cutting import time from **4 minutes to 45 seconds** through batched RabbitMQ processing and SHA-256 deduplication, with S3-triggered ClamAV Lambda scans blocking ~120 infected packages.
- Delivered AI question-variant generation using Angular, .NET, RabbitMQ, and Python on AWS Lambda, **cutting authoring time by 40%** and generating contextual and numerical variants 3× faster during the back-to-school peak, with SymPy validation, prompt sanitization, progress tracking, and template locking.
- Architected a hybrid search and RAG pipeline using Elasticsearch and Qdrant, cutting retrieval latency by one-third and **lifting question-classification accuracy 25%**, while generating metadata for 10K+ documents and powering context-aware retrieval across the platform.
- Unified 3+ fragmented grading SDKs into a single library supporting **12 question types across 4 teams**, removing ~60% of duplicated evaluation logic and cutting new-engineer onboarding 30% through one shared grading interface.
- Resolved **200+ production issues** via root-cause analysis, cutting mean time to resolution 20%, and owned release-impact communication and escalations across support, product, and development in Agile delivery.
- Led engineering execution across 5+ concurrent product deliverables, driving sprint planning, **mentoring 4 developers**, and resolving cross-team dependencies to deliver releases on time, improving delivery predictability 15%, while contributing production code.

N R Crew Jan 2023 – Dec 2023
Software Developer *Ahmedabad, India*

- Co-developed Tenderflow, a full-stack tendering platform on ASP.NET Core, Angular and PostgreSQL with Flyway-versioned migrations, replacing spreadsheet-based quoting with reusable templates and automated cost estimation, **cutting quote turnaround time 50%** and keeping pricing consistent across environments.
- Developed a .NET digital asset library on App Service and Blob Storage for **5,000+ assets**, with metadata tagging, background thumbnail and ZIP generation, and short-lived SAS download URLs, cutting lookup time 2×.
- Implemented a pricing engine for material and labour costs, handling markups, unit conversions, and rounding rules to keep estimates accurate and consistent, eliminating **~70% of manual pricing errors**.
- Improved dashboard performance by optimizing PostgreSQL queries with batched data retrieval and database-level aggregations, reducing unnecessary data processing and **cutting dashboard load time 45%**.
- Standardized CI/CD across teams with reusable GitHub Actions workflows, Docker and Azure Key Vault-managed secrets, **reducing deployment time 83% (35 minutes to 6 minutes)** and supporting 5+ daily releases.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, SQL, Python, HTML/CSS

Frontend: Angular, RxJS, Angular Material, micro-frontends, WCAG accessibility

Backend: .NET 8/10, ASP.NET Core, REST APIs, SignalR, RabbitMQ, Redis, Entity Framework Core, LINQ, Dapper

Databases: MySQL, PostgreSQL, pgvector, Qdrant, MongoDB, Elasticsearch, Flyway

Cloud & DevOps: Azure (App Service, Functions, Blob Storage, Key Vault), AWS (Lambda, ECS Fargate, DynamoDB, EC2, S3, SQS, RDS, CloudWatch), Docker, GitHub Actions, CI/CD, Git, Jira, Agile/Scrum

Testing & AI: xUnit, pytest, Testcontainers, Vitest, RAG, vector search, SymPy, OpenAI API, Azure Speech SDK

EDUCATION

Government Engineering College, Gandhinagar Gandhinagar, India
B.E. in Computer Engineering, CGPA 8.15/10 *2020 – 2024*